

Bhanuja Karumuru

📍 New York, USA 📞 +1 3474459258 🔗 [linkedin.com/in/bhanujakarumuru](https://www.linkedin.com/in/bhanujakarumuru) ✉ bk3170@nyu.edu

EDUCATION

New York University, MS in Computer Engineering New York, USA | Aug 2025 – May 2027
Relevant Coursework: Data Center and Cloud Computing, Parallel and High-Performance Computing, Internet Architecture and Protocols
NIT Sikkim, B.Tech in Electronics and Communication Engineering Sikkim, India | Dec 2020 – May 2024
Relevant Coursework: Computer Architecture, Computer Networks, Machine Learning, Neural Networks, Nature-Inspired Computing

TECHNICAL SKILLS

Programming & Systems: Python, C/C++ , Java, JavaScript, SQL, Bash, CUDA, Git, Linux
Core CS: Data Structures & Algorithms, Object-Oriented Design, System Design, Concurrency, Multithreading, REST API Design
Backend Frameworks: Flask, Django, FastAPI, PostgreSQL, MongoDB, Supabase, Microservices, Redis
Cloud DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD
AI/ML Data Science: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, Pandas, NumPy, pytest

WORK EXPERIENCE

Research Contributor, Secure Systems Lab, New York University New York, USA | Sep 2025 – Present
LIND Project – Virtualization-Based OS Security for ML Systems

- Engineered kernel-level isolation mechanisms and hypervisor-level privilege controls for a Linux security framework, reducing cross-VM attack surfaces in multi-tenant ML execution environments.
- Implemented namespace confinement and resource sandboxing primitives in C, improving workload isolation and enabling secure high-throughput execution across virtualized environments.

Software Development Engineer, SalesUp, Yellowlake Technologies Services Private Limited Kolkata, India | July 2024 – July 2025
(Promoted from Software Engineering Intern)

SalesUp CRM Platform – Modular Backend Architecture & Deployment Automation

- Designed and implemented asynchronous task processing pipelines and REST APIs for a high-traffic CRM platform serving concurrent users, improving system throughput by 40% and reducing P99 latency by 25%.
- Built an LLM-powered content automation backend with vector search and caching layers, reducing manual campaign workload by 40% and improving multi-channel delivery efficiency by 25%.
- Engineered containerized microservices with automated CI/CD using Docker and Jenkins, enabling zero-downtime deployments and reducing release cycles by 30%.

PROJECTS

Distributed Log Aggregation System

- Built in Python using producer-consumer concurrency and B-tree indexing for low-latency real-time search across containerized microservices.
- Designed REST APIs for log ingestion, querying, and filtering with caching layers, achieving sub-50ms query response times under high-throughput writes.
- Deployed on AWS EC2 with Docker, enabling horizontal scaling and fault-tolerant operations; validated end-to-end correctness with pytest unit and integration tests.

Microservice-Based Task Orchestration Platform

- Architected a task orchestration platform handling 10,000+ daily background jobs with priority queuing, retry logic, and audit logging.
- Designed async task queues, caching layers, and REST endpoints using Flask and Supabase for reliable distributed execution.
- Deployed Dockerized services on AWS EC2 with GitLab CI/CD, enabling automated builds, zero-downtime rollouts, and horizontal scalability.

End-to-End Fraud Detection Platform

- Engineered a high-throughput pipeline processing 1M+ daily transactions with automated ETL, containerized microservices on AWS EC2, and monitoring and rollback support.
- Built daily model retraining, continuous deployment pipelines with offline validation, ensuring minimal downtime & reliable versioned updates.
- Trained ensemble models (logistic regression, random forest, gradient boosting) achieving AUC 0.96 and reducing false positives by 35% through feature engineering.

Distributed Rate Limiter

- Implemented a distributed rate limiter in Python supporting Token Bucket and Sliding Window Log algorithms with thread-safe concurrent request handling via atomic operations.
- Built a Redis-backed shared state layer enforcing consistent rate limits across service instances with sub-5ms overhead under high concurrency.
- Validated correctness and thread-safety with pytest unit and integration tests simulating burst traffic across concurrent clients.

PUBLICATIONS

In-Domain Data Augmentation for Dysarthria Severity Classification, IEEE SPCOM 2024
Proposed in-domain augmentation techniques for dysarthric speech under low-resource conditions, improving severity classification accuracy; accepted at IEEE SPCOM 2024.